

Hihi A Cheater Teacher Report:

By Liam Twigley, Sebastian Allan, Erica Kim, Song Ming Ten & Qixiang Yang

Introduction:

This report provides the solutions to two tasks, as part of the Cheater Teacher project. These tasks include:

1. Identifying possible cheating teachers in a large dataset through characteristic identification methods using R code.
2. Formulating a probability model to estimate the proportion of results from cheater teachers and identify how they cheat.

The datasets used include:

1. **ChildrenScores:** Large dataset comprising student data, including mean grade scores and scores from year to year beginning at Y3 to Y7 with around 1000 unique teachers for each year level.

Task One: Find the Cheats

Objective:

The primary objective of this task was to develop a method for identifying possible cheater-teachers within the dataset. Manually identifying cheaters would be both inaccurate and time-consuming due to the data's magnitude. Therefore, we employed statistical techniques using R to automate this process.

Methodology:

The first step involved calculating the difference between yearly scores and mean scores for each student and storing it as a new variable in our dataset. We did this through the **mutate** function in the **tidyverse** package by subtracting the student scores by the mean scores i.e. **mutate (diffY3 = scoreY3 - meanScore**, this provided a metric to identify unusual increases in scores across all year levels.

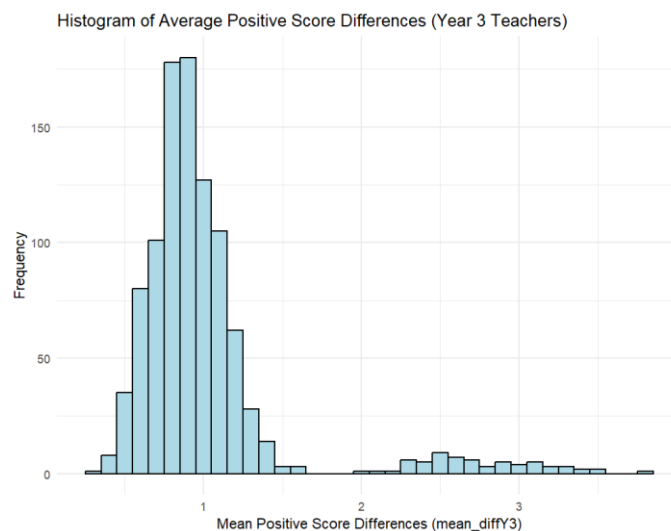
Our second step includes filtering the differences to strictly positive differences. Since the goal of cheating teachers is to increase the proportion of students passing, we focused on positive increases in scores. Using the **case_when()** function, we filtered out non-positive differences.
i.e. **mutate(pdifffY3 = case_when(diffY3 > 0 ~ diffY3, T ~ 0).**

The third step in our method is calculating the mean of positive differences for each teacher in every year (y3-y7). For each teacher, we calculated the mean of positive differences for every year (Y3-Y7) using `group_by()` and `summarise()`. i.e.

`summarise(mean_diffY3 = mean(pdifY3))` this then stores the values in a new data frame called **`data_Yx`** where x is the corresponding year and can be used in our next step.

The fourth step includes filtering out teachers that are not suspicious. Using the **`filter`** function any teacher with a positive mean difference of less than or equal to 1.9 would be removed and flagged as a non-cheater i.e. **`filter(mean_diffY3 > 1.9)`**. This was to further analyze suspicious teachers and eliminate non-suspicious activity. This was then repeated every year in our dataset.

The histogram of average positive score differences indicates that most teachers had a mean positive difference between 0.5 and 1.5. However, a distinct group had a mean positive difference above this range. Teachers with a mean positive difference exceeding 1.9 were flagged as potential cheaters, as their students' performance improvements were unusually high.

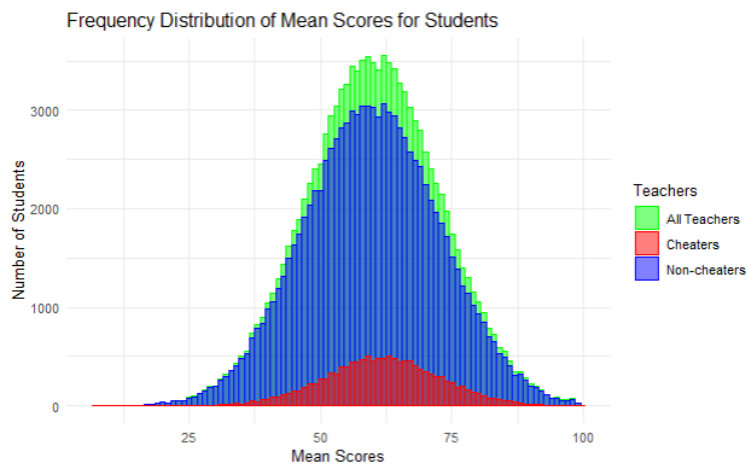


Our fifth and final step includes combining the suspected cheater teachers from the previous step into a single data frame and create a new csv comprised of a single column for CheatID and includes 186 rows of cheater teacher IDs. The method use can be seen as such:

```
final_cheater <- as.data.frame(c(data_Y3$teacherY3, data_Y4$teacherY4,
                                data_Y5$teacherY5, data_Y6$teacherY6,
                                data_Y7$teacherY7)) final_cheater <-
final_cheater %>% rename(CheatID = 1)
```

Plots:

To get a better picture of how the score distributions vary from each of the different group's cheater teachers and non-cheater teachers we created a histogram graph. What we ended up seeing is that the cheaters (red) tend to have a higher overall proportion of students scoring above 50% (passing) over the baseline group, and the non-cheaters along with higher overall mean scores. This indicates that the cheater teachers had a different distribution of scores to the non-cheater teachers and clearly used methods to increase their overall student performance in non-legitimate ways.



Results:

We calculated the overall percentage of cheater teachers in the dataset by dividing the number of flagged cheater teachers by the total number of unique teachers:

`cheater_percentage <- (flagged_cheater_teachers / total_unique_teachers) * 100.`

This revealed a cheater prevalence rate of approximately **7.47%**.

Task Two: How do they cheat?

Objective:

The objective of Task 2 is to formulate a probability model to separate cheat-scores from noncheat-scores within the classes of cheater teachers. The goal is to understand the cheating mechanism by analyzing the distribution of scores and estimating the parameters of the probability model.

Model Formulation:

In Task Two, our primary goal was to distinguish between the impacts of cheating and non-cheating teachers on student performance by cleaning the data and calculating residuals to analyze these effects.

1. Data Cleaning and Classification:

We started by reading and processing the ChildScore.csv file, separating teachers into two categories: cheaters and non-cheaters. Based on the **cheatID** in the Hihi-A-Cheaters.csv file, we identified whether each teacher was a cheater. Using **mutate** and **ifelse**, we processed the student scores from years Y3 to Y7. For cheating teachers, the student scores were set to NA, while the scores for non-cheating teachers were retained. Finally, we used **rowMeans()** to calculate new average scores for each student, enabling more accurate data analysis.

2. Residual Calculation:

After cleaning the data, we restructured the data frames and created a column for the residuals. First, we used **gather()** to convert the score columns into a long format, then calculated the residuals for each student by subtracting their actual score from the average score:

3. Mle Model:

For Our MLE method, we borrowed this from week 8; this was used to estimate the means and standard deviation, which are our parameters. The cheater and the non-cheater teacher classes are assumed to be normal distributions. We used this to help produce the parameters of our combined residual distribution.

Results:

Through residual calculation, we found that:

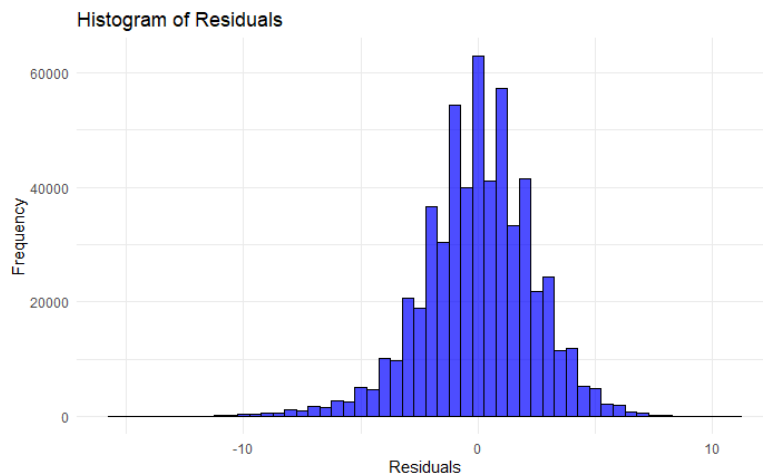
Cheating Teachers: Residuals were significantly positive, indicating that student scores were influenced by dishonest practices.

Non-Cheating Teachers: Residuals were close to zero, showing that student performance was more aligned with expectations.

Plots:

Next, to visualize the distribution of residuals, we used the ggplot2 library to create histograms of the **residuals**. These histograms clearly display the difference in residual distribution between the students taught by cheating and non-cheating teachers. Students under cheating teachers generally had larger positive residuals, indicating that their scores were artificially inflated, while the residuals for non-cheating teachers were closer to zero, showing more natural performance.

```
MLE estimates for the mixture of two normal distributions (combined residuals):
Mixing proportion (q): 0.9999147
Mean of first normal (u1): -0.0005869692
Standard deviation of first normal (sigma1): 2.356449
Mean of second normal (u2): 4.999291
Standard deviation of second normal (sigma2): 4.35801
```



Conclusion:

In Task Two, we successfully built a model to distinguish between the impact of cheating and non-cheating teachers on student performance by cleaning the score data and calculating residuals. The data cleaning and residual analysis showed that students under cheating teachers had significantly inflated scores, while students under non-cheating teachers displayed more natural performance. The visualization further confirmed the difference in the residual distribution between the two groups. Task Two is closely related to Task One, where we identified cheating teachers using statistical methods. Task Two validated how these teachers affected student scores, providing further insights into the manipulation of scores by cheating teachers.